

What the Data Says, and What Singapore Taught Me

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This study was commissioned by

Dr. Yinping Yang

in her capacity as CATOS Director.

The report is dedicated to her memory, with the hope that it caught some of her vision and her commitment to understanding online harm in this ecosystem.

She asked the questions. What follows is my attempt at answers.

The arc of this talk

One corpus, five findings, and the part I hope travels

1

The setting

10 min

154M posts · 3 platforms
6½ years · no identities

2

The findings

12 min

Five results, each with
the test that bounds it

3

Worked case

6 min

GE2025 end to end —
and
the mistake we caught

4

The lessons

7 min

How to measure an
environment honestly

One thread throughout: **every number here is a floor**, and the discipline that makes it a floor is exactly what makes it **usable**.



THE BRIEF • 2025–2026

**I was asked what is
actually out there.**

Not what we fear is out there, and not what one year happened to show

01

Four questions, asked in order of narrowing

The CATOS Dataset — the time-extended successor to a corpus covering only 2025

01

How much
coordination?

And what infrastructure
carries it

02

Which harms, how
large?

Scams, hostility, and
where
safety systems fail

03

What changed over
time?

Whether one year
guides
the next

04

What does it
look like?

One event, detection
through validation

The premise I was asked to test: that extending the corpus **in time** was not a quantitative convenience but a **qualitative** one.

It was. **Three conclusions** a 2025-only study would have supported are overturned by the full span.

What the year produced

Two deliverables, and three things that keep working after the contract ends

~150

pages · 17 chapters
6 appendices

1

preprint
arXiv:2606.21996

5

briefings delivered
DSO · DSTA · MDDI · IMDA · OTS
Forum

1

grant call reopened
OTS Call 3, AI-safety frontier

The report and the preprint are the deliverables.

Every finding is chained to the test that bounds it, and Chapter 15 lists what I looked for and **could not find**.

The rest is infrastructure.

A reproducibility manifest so this re-runs without me. A revised grant call opening agentic and multimodal AI safety. Introductions to MDDI, IMDA, SUTD and MLCommons left open and warm.

THE CORPUS

**154 million posts, and
no idea who wrote them.**

One constraint shapes every sentence in the report

02

What the CATOS Dataset is

Facebook, Reddit and YouTube · January 2020 – June 2026

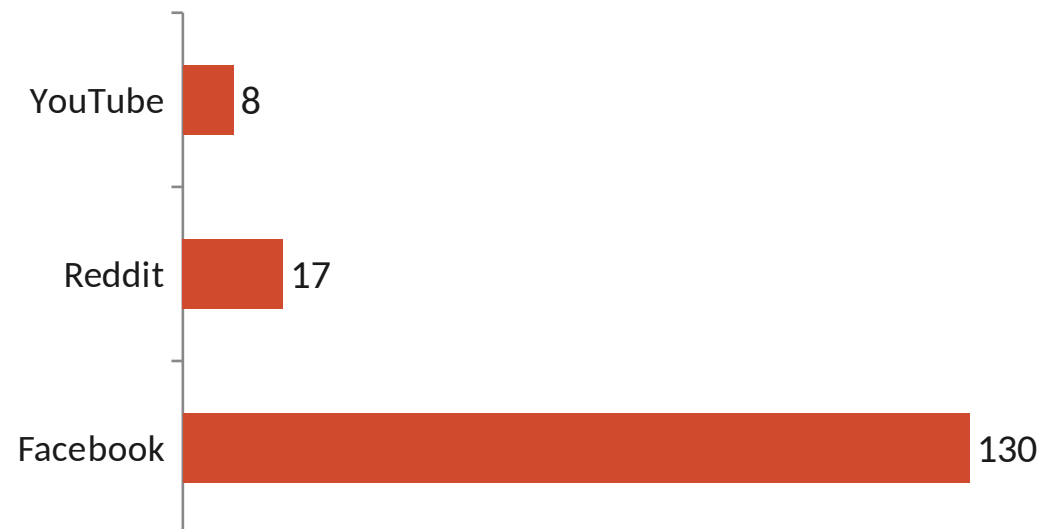
154,284,618 items across three platforms

267 distinct spaces — pages, subreddits, channels

11.96M URLs pointing to **105,022 domains**

One national conversation running in English, Malay, Mandarin and other Chinese varieties, and Tamil — and out into the messaging apps people are funnelled toward.

Corpus composition · millions of items



84% of everything here is Facebook — the one platform with no provider safety score at all. Hold that thought.

The constraint that shaped everything

F000 · the dataset carries no usable account identities

WHAT WE CAN SEE

What was said

When — to the second

Where — page, subreddit, channel

What it linked to — domains, lifespans

WHAT IS PERMANENTLY OUT OF REACH

Who said it

Sockpuppet or bot-network mapping

Attribution to any actor or state

Whether repetition was authentic

The account-level toolkit platform integrity is built on is **out of scope by construction**.

So the study was rebuilt around **content, timing and infrastructure** — what the data can carry.

So when this report says “coordinated”

It means one thing, and the room should hear only that thing

The same text, verbatim, reposted across several distinct communities, in a temporal pattern **tighter than chance predicts.**

Read it as “verbatim-repeated across spaces” — and no more. Equally consistent with:

An organised campaign

The reading people jump to.
It is one of four.

Many people, loosely aligned

Independently favouring the same slogan.

One viral post

Quoted, screenshotted, re-pasted at scale.

Platform mechanics

Share buttons, cross-posting, moderation bots.

This report identifies no one as responsible for anything. Crossing that line, even under pressure for a more dramatic finding, forfeits the trust that makes the rest usable.



THE FINDINGS

Five results I would defend anywhere.

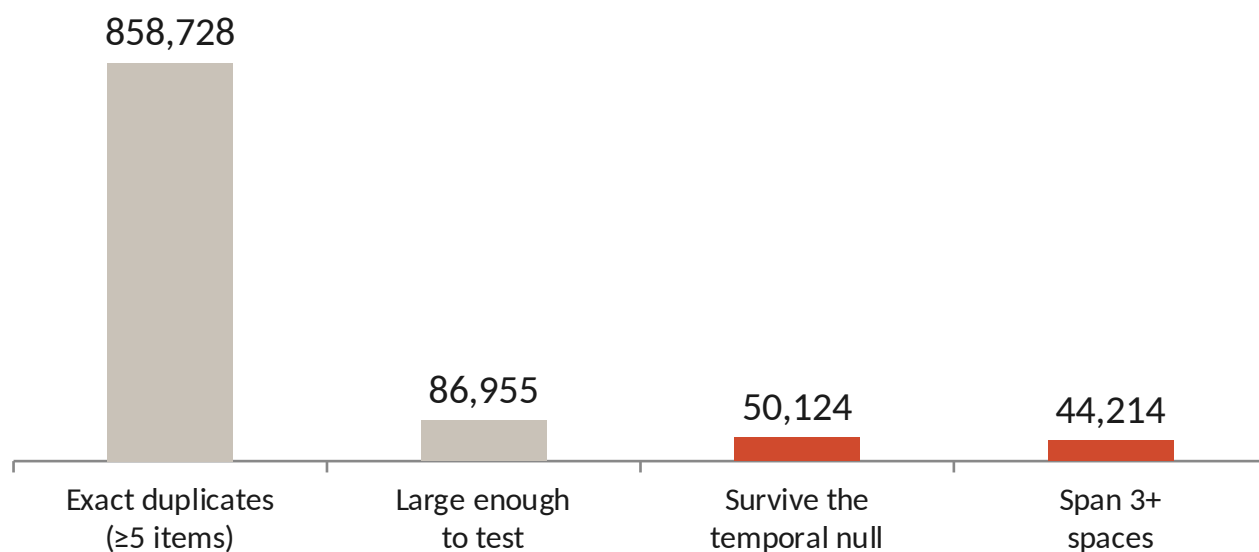
Each one paired with the specific test that bounds it

03

Coordination is real, and substantial

Chapter 4 • exact-duplicate clustering, then two filters that do real work

From raw duplicates to policy-relevant coordination



The permutation null removed **42%** of flagged clusters.

Raw duplicate-counting would have overstated coordination by nearly half.

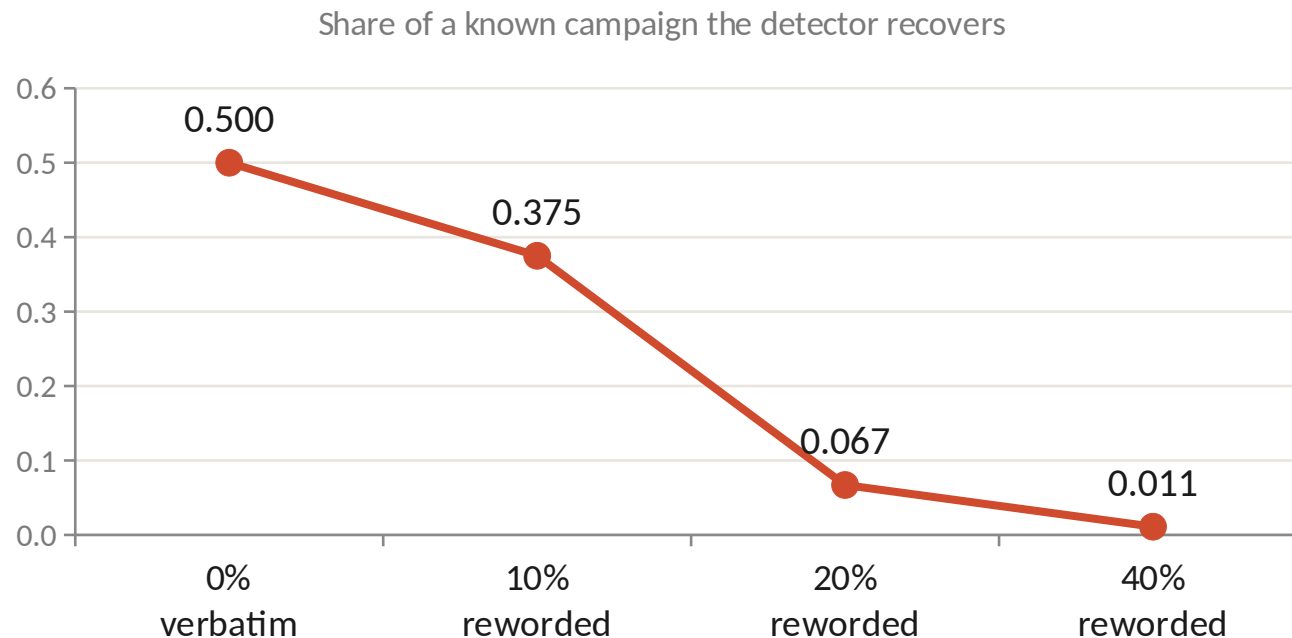
44,214 crossed three or more community boundaries.

The only measure that means anything for platform integrity.

Largest single cluster: **239,336 items**, one verbatim string.

...and every count is a floor

We injected known campaigns into the real corpus, then reworted them



A campaign that varies its wording even slightly is **almost entirely invisible**.

No count here estimates true coordination. Each is a high-precision floor, and the truth is very likely much larger.

Any comparison across time or platform compares **verbatim floors**. A group that paraphrases more merely looks less coordinated.

This is the slide that makes the report honest — **and it is also the strongest funding case in it.**

The dominant harm is commercial

Chapter 6 • precision-corrected floors, never raw cue-hits

769,601

raw scam cue-hits
before correction

× 0.208

measured gold-set
precision

~160,307

the floor I actually
report

5.6×

enrichment over
the corpus base rate

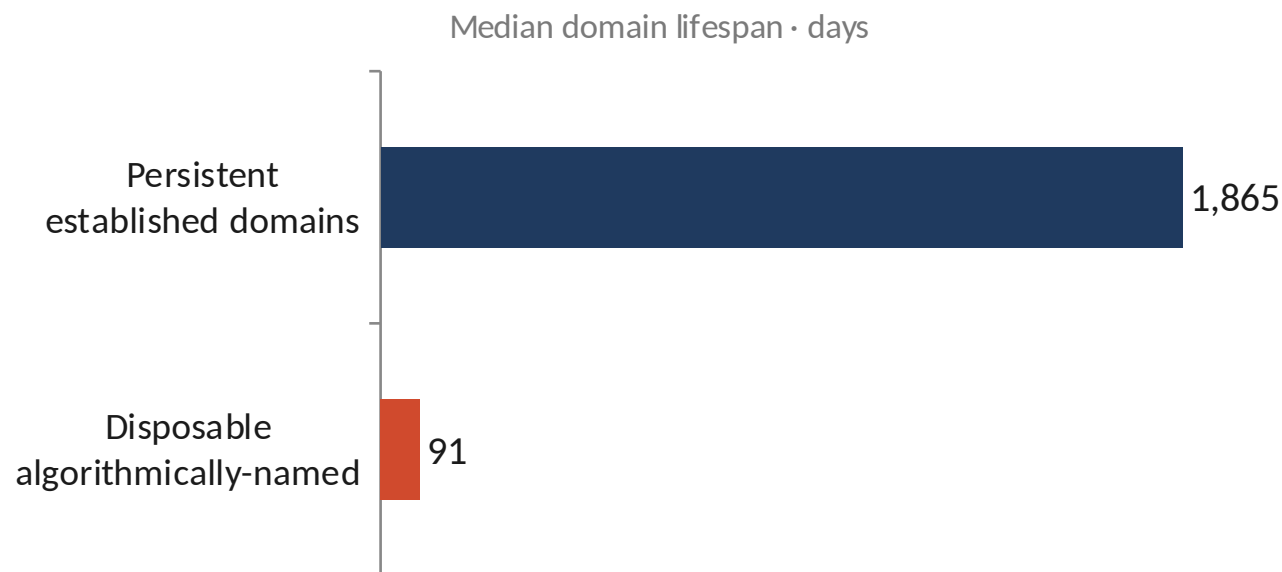
Fake investments • job offers • giveaway impersonation • gambling • recovery scams

All the same shape: a **polite, transactional** message carrying a link that funnels the reader **off-platform into a private messaging app**, where nothing is observable at all.

Politics is what everyone asks about. Fraud is what the data is full of.

Two infrastructures, two lifecycles

Chapter 5 • the signal that survives paraphrase



A 20-fold gap — and **78.6%** of tested burner domains were pushed in synchronised bursts.

Infrastructure survives rewording.
Paraphrase defeats the text detector; it does not conjure a new domain.

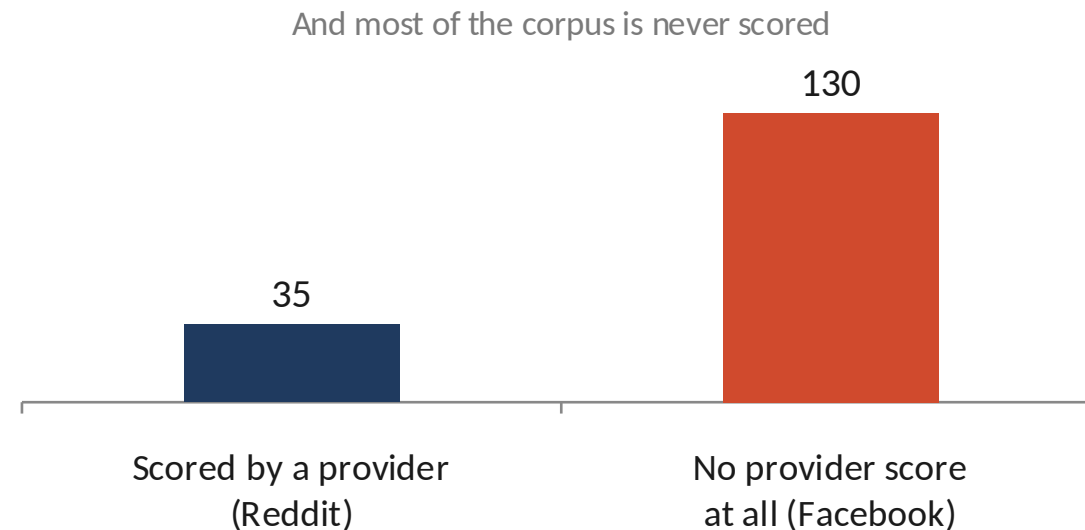
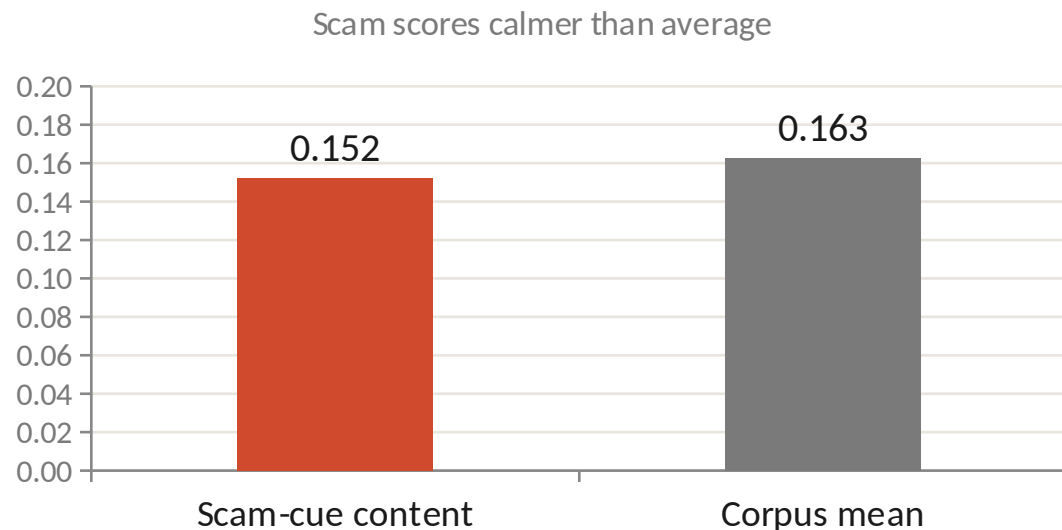
Visible from **public domain data alone**, with no platform cooperation.

But the flag misfires on **1 in 10** oddly-named legitimate sites. A candidate generator, not a verdict.

A 91-day cycle is **invisible in any single-year window** — which is the next finding, arriving early.

The safety systems cannot see the main harm

Chapter 7 • a negative result about a common instrument



Scam content is **polite by design**. It is optimised to pass exactly the classifier we trust to catch harm. This is not a coverage gap that more compute closes. A safety system tuned to toxicity, harassment and hate will **systematically under-serve** a market where commercial fraud is the principal measurable harm.

The two blind spots are mirror images

Each layer is blind roughly where the other is strong

Coordination & infrastructure

Provider toxicity scoring

Scam & funnel

Strongest lift in the toolkit

Weakest — scores below average

Xenophobia

Near-organic, little structure

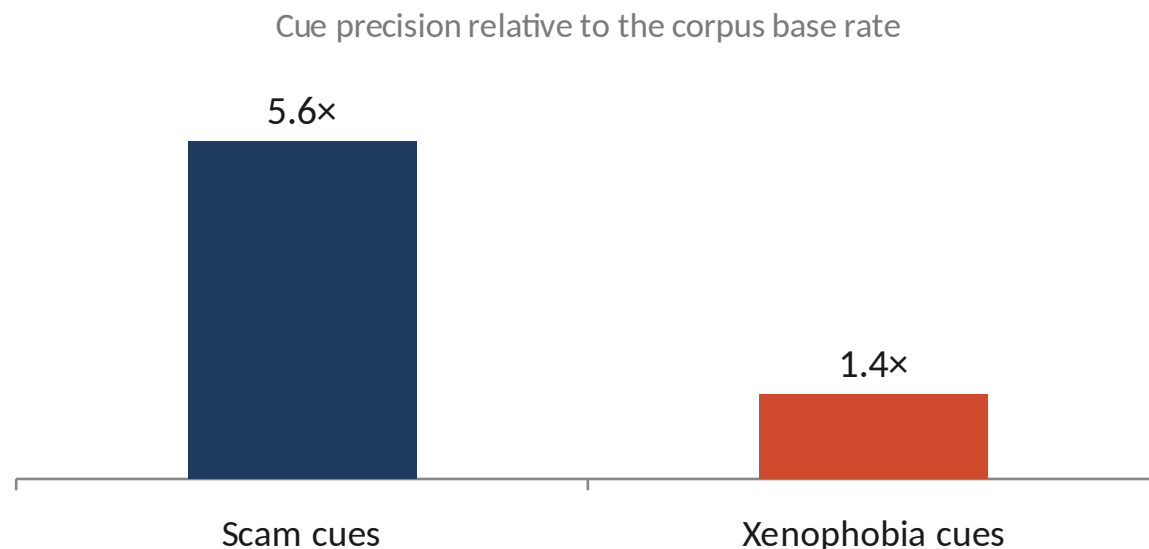
Comparatively well flagged

Neither layer alone gives a complete picture of either harm.

That is an argument for **combining** them — not for choosing between them.

Xenophobia is pervasive, not concentrated

A genuine negative result, and the least comfortable one to deliver



Xenophobia cue precision is **high** — 87%.

But so is its base rate: roughly **62%** of definite labels in a random sample.

High precision on a common phenomenon is not a detector. It is a thermometer.

The signal is not a needle in a haystack. It is a property of much of the hay — and you cannot address that by taking down a network, because there is **no network to take down**.

A coordinated-campaign result would have been easier to deliver. It comes with an obvious intervention. This one does not, and saying so is part of the job.

A single year can invert the truth

The methodological spine • three conclusions a 2025-only corpus would support

Commercial spam looks near-organic

0.19

2025 funnel lift



15.8

2020 funnel lift

Scale inverted

Sharing looks platform-balanced

parity

in 2025



FB-led

in 2020

Direction inverted

Disposable infrastructure looks absent

91

day burner median



1,865

day persistent median

Invisible in one year

Longitudinal span is not more data. It is a correction mechanism.



THE WORKED CASE • GE2025

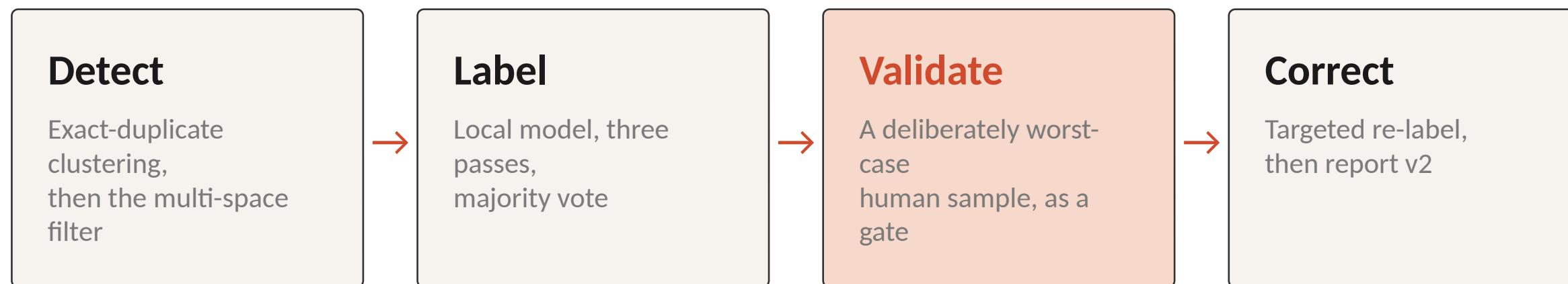
One election, start to finish.

Including the mistake the pipeline caught before I published it

04

The pipeline, on one well-understood event

Chapter 14 · the same four steps used everywhere else in the report



Political cypypasta peaks in the **week before Polling Day**, then collapses.

The third box is the one people skip. It is the one that saved this chapter.

4.5%

mean coordination share
across the corpus

22.0%

raw peak, April 2025
— read the next slide first

0.98

mean self-reported
model confidence

What the validation gate caught

A 3-billion-parameter model, confidently wrong in exactly one place

VERSION 1 • AS THE MODEL REPORTED IT

42 clusters • 1,925 items

labelled anti-opposition

Reliable on **direct** pro/anti stance (80–100%)

Agreed with itself across all three passes

WHAT A HUMAN REVIEWER FOUND

16 of 42 mislabelled — 38%

Sarcastic anti-incumbent posts read as anti-opposition

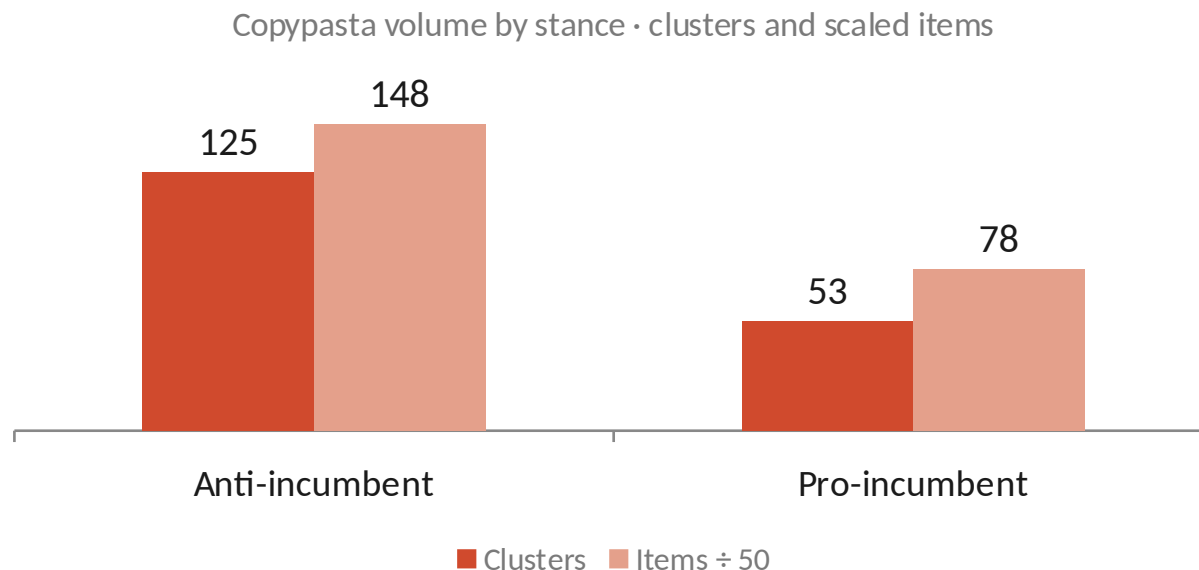
Neutral structural commentary swept in with them

Ten of the sixteen carried **no partisan valence at all**

Small-model labels are hypotheses, not measurements. Self-consistency is not correctness — it was most confident exactly where it was wrong.

The validated headline

Corrected v2 — the numbers I actually report



2.4 : 1 by cluster count · **1.9 : 1** by item count

Pro-incumbent copypasta is a small set of **high-reach slogans** (73 items per cluster). Anti-incumbent is a wider variety of **longer critiques** (59 per cluster).

And what it is not: evidence of coordinated inauthentic behaviour, foreign interference, or manipulation.

High-volume criticism of a long-governing party at its most exposed moment is **unremarkable** in a democracy. It is visible here only because some of it took verbatim, cross-space form.

And the same week's largest raw cluster — **13,014 items** — was Reddit AutoModerator boilerplate in a single space. **Check spread, never size.**



THE PREPRINT

Hate has a different shape in each language.

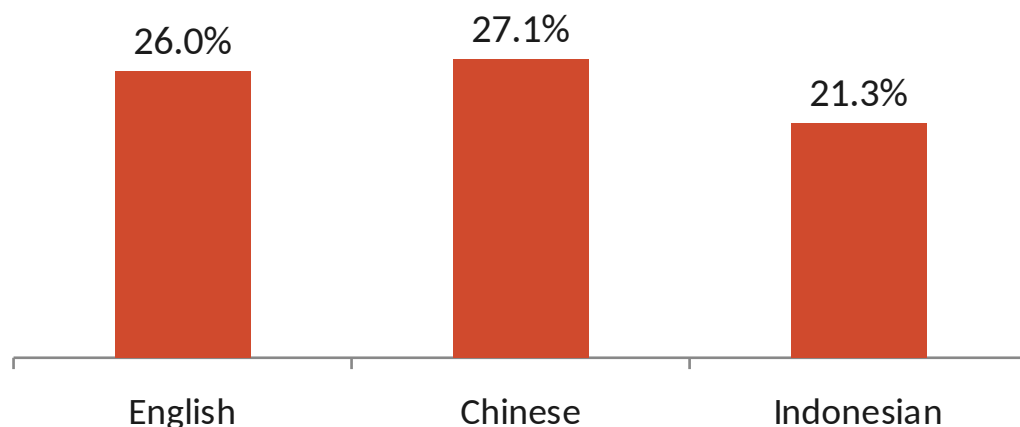
A cross-lingual audit — the one piece of this work that is public

05

Cultural targets, structural frames

8,561 candidates adjudicated · 2,190 confirmed hostile · three languages

Share of sampled candidates confirmed hostile



Targets are language-specific. English hostility spreads broadly; Chinese concentrates on a **Malay-Chinese divide**; Indonesian concentrates on **religion**.

Frames are determined by the target's category, and are shared across communities.

Resonance rewards nativist hostility over religious hostility.

Morals lean in-group-protective — but under a labelling confound.

Three of the four lenses replicate on the full 2020–2026 record. The practical consequence: English-only monitoring systematically misses the Chinese- and Indonesian-language fault lines, because they target different groups entirely.



THE LESSONS • WHAT TRAVELS

What Singapore taught me.

The dataset stays behind. This is the part I hope does not.

06

Three gates, and what each one caught

None of them expensive. All three changed a headline.

1

Calibrate against a null

Every detector gets a randomisation baseline and a sensitivity curve before its output is believed.

Removed **42%** of flagged clusters, and reframed every count as a floor

2

Correct by measured precision

Every raw count is multiplied through a gold-set precision estimate and reported as an explicit floor.

Turned 769,601 cue-hits into a defensible **160,307**

3

Validate against worst cases

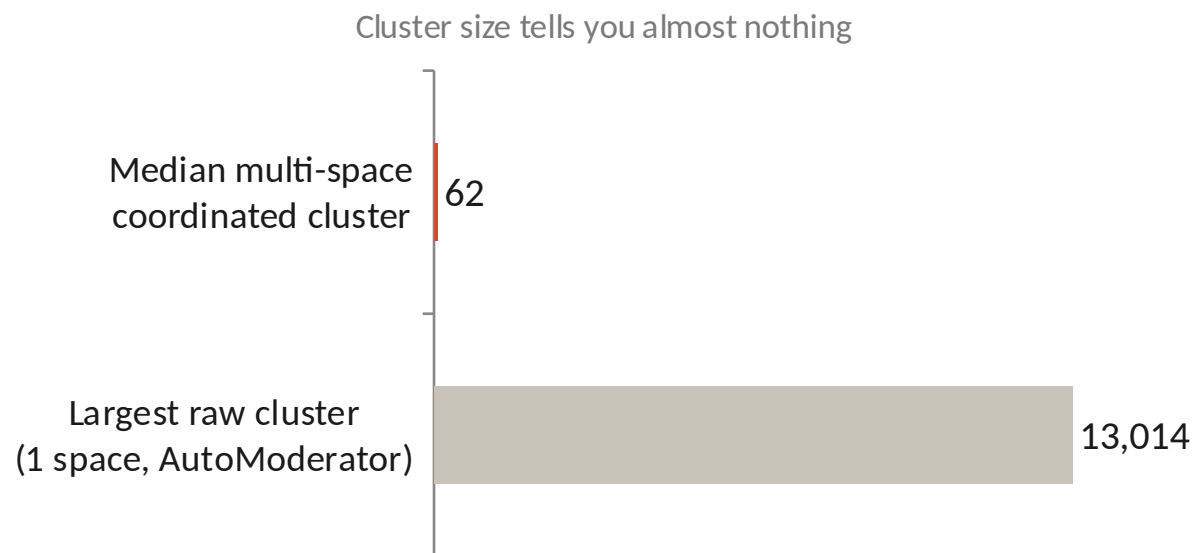
Every model label is checked against a deliberately adversarial human sample before it is reported.

Caught a **38%** mislabel rate hiding behind 0.98 confidence

Each gate caught a real error that would otherwise have reached the headline.

Trust spread, not size

The biggest spike in six years was a moderation bot talking to itself



April 2025 shows a raw coordination share of **22%** — the highest month in the corpus.

Event-anchoring traces the largest cluster to **one space**: Reddit's AutoModerator, posting rule text that is verbatim by construction.

The trustworthy measure is the **multi-space fraction**. Raw size is a trap.

Five things I'd tell whoever picks this up

Not report language — what I actually say to students

1 Report floors, and say so every time

A floor everyone understands beats an estimate nobody defends

2 Never use toxicity as a scam screen

The largest harm here is polite. It is engineered to pass

3 Assume the model is confidently wrong somewhere

Sarcasm was the where. Next time it will be something else

4 Measure more than one year

Two of three headline directions flipped at six years

5 Publish the negative results

A report that only says what it found is not trustworthy

The negative-results chapter is the one I'd read first if someone handed me this report. It is Chapter 15.

What I would fund next, in order of leverage

Each gap is a specific, fundable capability — not a wish list

1 Paraphrase-aware detection

Turns every floor into a bounded estimate

HIGHEST LEVERAGE

2 From floors to prevalence

Annotation work, not new collection

CHEAPEST WIN

3 A multi-model labelling stack

Failure modes published, not just accuracies

HIGHEST RIGOUR

4 Instagram, TikTok, X

Delivered but unanalysed; lead-lag is moving

WIDEST COVERAGE

5 Infrastructure-anchored monitoring

A 91-day burner cycle needs a standing monitor

MOST OPERATIONAL

Recall first, then calibration, then labelling rigour, then coverage, then timeliness. Each arrives paired with the validation that bounds it — or it should not arrive at all.

I came looking for the wrong thing

And the data spent six months telling me so

WHAT I ARRIVED EXPECTING

Influence operations.

Political manipulation, coordinated networks, an election to be defended.

Fifteen years of my field points there.

WHAT THE DATA KEPT SAYING

Fraud, infrastructure, language.

The biggest harm is commercial.

The durable signal is a domain, not a sentence.

Fault lines differ **in every language.**

The politics was real, bounded, and episodic. It was not the story.

The correction that mattered most

It did not come from a model

I had the corpus down as Bahasa Indonesia. **A colleague read the draft carefully and told me I was wrong — it is overwhelmingly Malay.**

That single correction changed a chapter, and it changed what the hate audit is allowed to claim.

No amount of compute would have caught it. **A research centre would.**

That is what CATOS was for me: a place where somebody knew the language, argued with my thresholds, checked my base rates, and told me when I had it wrong — early enough that it never reached print.

Three takeaways

If everything else falls out of memory on the way to the lift

1

**The harm is
commercial**

scam beats politics, by volume

and it is polite enough that toxicity
classifiers score it below average

2

Every count is a floor

and saying so is the point

a floor everyone understands is
worth more than an estimate
nobody defends

3

**One year can invert the
truth**

**measure long, or say you did
not**

two of three headline directions
flipped once six years were
available



Thank you

Emilio Ferrara • emiliofe@usc.edu

For the data, the arguments, the corrections, and the year.
And for Dr. Yinping Yang, who asked the questions this work tries to answer.

HANDED OVER

Technical report

Preprint • arXiv:2606.21996

Reproducibility manifest

Every limitation, numbered

The corpus and the coordination funnel

For questions · every count is a floor unless stated otherwise

154,284,618

items, Jan 2020 – Jun 2026

129,506,705

Facebook comments (84%)

16,487,473

Reddit posts and comments

8,290,440

YouTube items

858,728

exact-duplicate clusters, ≥ 5 items

50,124

survive the null (57.6% of 86,955)

44,214

span 3 or more distinct spaces

239,336

items in the largest single cluster

267

distinct spaces

11,959,097

URLs

105,022

distinct domains

2.21

median effect size among survivors

Detector recall across rewording: $0.500 \rightarrow 0.375 \rightarrow 0.067 \rightarrow 0.011$ at 0 / 10 / 20 / 40 percent.

Harm, infrastructure and time

The rest of the numbers

~160,307

scam floor
769,601 cue-hits × 0.208 precision

~256,573

xenophobia floor
295,048 cue-hits × 0.870 precision

5.6× / 1.4×

enrichment
scam vs xenophobia

0.152 / 0.163

scam toxicity vs corpus mean
provider scoring, Reddit only

91 / 1,865

burner vs persistent domain
median lifespan, days

78.6%

DGA synchrony
55 of 70 domains at $q < 0.05$

15.8 → 0.19

funnel lift
2020 → 2025

4.5% / 22.0%

coordination share
mean / raw peak (Apr 2025)

2,190

confirmed hostile
of 8,561 adjudicated

What this study cannot do

Chapter 15 · stated so that no finding is over-read

No attribution

No actor, group or state is identified. Structural, not cautious.

No prevalence

Counts are floors, never population rates.

No paraphrased coordination

Recall falls to 0.011 at 40% rewording.

No inauthenticity claim

Repetition is measured. Intent and automation are not.

Partial platform coverage

Instagram, TikTok and X delivered but unanalysed.

Screening is not incidence

Safety-screen indicators have unknown precision.

Significance thresholds ($q < 0.05$, Benjamini–Hochberg) control the false-discovery rate across a tested population. They do not certify any individual cluster, URL or domain.